

Illustrative Example of How to Think About the Outcome of Qubits Based on Their Classification

In this example, p is the probability that one experimental shot satisfies a predefined success criterion. It is separate from gate fidelity, readout fidelity, state fidelity, circuit fidelity, and logical error rate.

Model	Target p	Shots	Successes	Failures	Success Fraction (Observed p)	Wilson CI 95% Low	Wilson CI 95% High	CI Width (95%)	Required Shots for $\pm 2\%$	Required Shots for $\pm 1\%$	Required Shots for $\pm 0.5\%$	Illustrative Cost (USD)
Toy working qubit	0.50	10,000	5,025	4,975	0.5025	0.4927	0.5123	0.0196	2,401	9,604	38,415	\$0.60
NISQ qubit	0.60	10,000	6,010	3,990	0.6010	0.5914	0.6106	0.0192	2,305	9,220	36,879	\$0.60
Near-term qubit	0.70	10,000	7,008	2,992	0.7008	0.6917	0.7097	0.0179	2,017	8,068	32,269	\$0.60
Advanced near-term qubit	0.80	10,000	8,006	1,994	0.8006	0.7927	0.8083	0.0157	1,537	6,147	24,586	\$0.60
Fault-tolerant logical qubit	0.90	10,000	9,013	987	0.9013	0.8953	0.9070	0.0117	865	3,458	13,830	\$0.60
Topologically protected qubit	0.98	10,000	9,797	203	0.9797	0.9767	0.9823	0.0055	189	753	3,012	\$0.60
Ideal qubit	1.00	10,000	10,000	0	1.0000	0.9996	1.0000	0.0004	1	1	1	\$0.60

Toy working qubit	A minimal demonstrator for which approximately half of the shots satisfy the selected success criterion.
NISQ qubit	A noisy physical qubit used in shallow circuits with limited coherence, imperfect gates, and limited error suppression.
Near-term qubit	An improved NISQ-era system using stronger calibration, error suppression, error mitigation, dynamic circuits, and application-oriented workflows. Full logical fault tolerance remains outside this regime.
Advanced near-term qubit	A transitional regime involving higher-fidelity physical qubits, repeated error detection, deeper circuits, small error-detected codes, or early logical-qubit demonstrations. Sustained scalable fault-tolerant operation has yet to be reached.
Fault-tolerant logical qubit	An error-corrected logical qubit supported by repeated syndrome extraction, active decoding, and logical operations whose error rates remain below the applicable fault-tolerance threshold.
Topologically protected qubit	A qubit whose encoding provides intrinsic suppression of selected local error mechanisms. Control, readout, poisoning, leakage, and residual nonlocal errors can still influence the measured success probability.
Ideal qubit	A theoretical abstraction with perfect preparation, control, coherence, gate execution, and measurement, giving deterministic success for the specified criterion.

The Wilson score interval is calculated from the observed successes and failures. The required-shot estimates use

$$n = \left\lceil \frac{z_{0.975}^2 p(1-p)}{\epsilon^2} \right\rceil, \quad z_{0.975} = 1.959964,$$

where ϵ is the requested absolute half-width.

The cost column assumes a fixed illustrative price of \$0.60 per 10,000 shots, equivalent to \$0.00006 per shot. The ideal-qubit row treats $p = 1$ as a theoretical axiom. Empirical validation of an unknown probability would require many observations despite recording zero failures.